Research Article

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Wet-cupping on calf muscles in polycystic ovary syndrome: a quasi-experimental study

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Abstract

Objectives: Current modalities for treating polycystic ovary syndrome (PCOS) are not curative and using them for a long period is associated with adverse effects. According to Persian Medicine recommendations, wet cupping on calf muscles can induce menstrual bleeding. In the present study, the effect of wet-cupping (traditional bleeding from capillary vessels) was assessed on menstrual cycles (as primary outcome), hirsutism, and laboratory manifestation of PCOS.

Methods: A quasi-experimental study was conducted from 2016/5/21 until 2017/5/20 on 66 PCOS women within the age range of 20–40 years and a menstrual interval of at least 60 days during the last year. Participants' calf muscles were cupped on day 26 of their last menstruation cycle following the Persian Medicine recommendations. They were followed for 12 weeks and a menstruation history and physical examination was done twice (4 and 12 weeks after wet cupping). Insulin Resistance (IR) and Free Androgen Index (FAI) were evaluated at baseline and end of the study.

Results: Wet-cupping on calf muscles significantly improved menstrual cycles' frequency 0.37(95% CI: 0.13, 0.51), p-value=0.001 and hirsutism after 4 and 12 weeks of intervention were −1.9 (95% CI: −2.5, −0.5), p-value<0.001 and −1.4(95% CI: −2.1, −0.8), p-value<0.001, respectively. While it was not significant in changing the IR and FAI. About 33% of participants suffered from mild temporary discomforts (which were resolved within a few minutes of rest) and 9% reported mild long-term side effects.

Conclusions: It is considered that wet-cupping on calf muscles can be propounded as an optional treatment of PCOS for those not willing to use chemical medication.

Keywords: insulin resistance; Persian Medicine; polycystic ovary syndrome (PCOS); wet-cupping

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Introduction

Polycystic Ovary Syndrome (PCOS) is the most common endocrinopathy disease as it affects 5–15% of women of reproductive age [1]. While the diagnostic criteria change over time, based on the American Association of Clinical Endocrinologists (AACE), Androgen Excess and PCOS Society (AES) criteria of PCOS include any two of the following three features: chronic anovulation, clinical or laboratory hyperandrogenism, and polycystic ovarian morphology [2]. Hyper-insulinemia plays a critical role in the pathology of PCOS [1] by increasing the androgen production from theca cells [3] and decreasing sex hormone-binding globulin (SHBG) in the liver [4]. There are no specific curative approaches for treating this multi-system disorder [5] and long-term use of chemical-hormonal agents may be associated with adverse effects [6]. As a result, PCOS-affected women may tend to use alternative and complementary medicine.

Avicenna, the most famous Persian physician (980–1037 AD), recommended wet cupping on calf muscles for the treatment of oligo/amenorrhea [7, 8] and relevant disorders such as lipid disorders and diabetes mellitus [8–11]

Wet cupping has been approved by the World Health Organization (WHO) as a form of bloodletting. It is done by pulling a vacuum cup on the surface of the body, along with the use of incisions, to remove capillary blood [12–14].

Despite the widespread use of wet cupping in East Asia and the Middle East, just few clinical trials have been conducted to assess its efficacy on PCOS. However, improvements in reproductive abnormalities following wet cupping have also been reported in some studies [15, 16].

The present study aims to evaluate the effects of wet-cupping on a menstrual interval, insulin resistance, and hyperandrogenism in women with PCOS having Rotterdam criteria.

Materials and methods

Since we could not design a placebo procedure like wet cupping, this quasi-clinical non-randomized study was conducted before and after trial on 66 women with Polycystic ovary syndrome aged 20–40 years who were admitted to *Fatemieh* gynecology hospital in Hamadan city, Iran.

Oligomenorrhea is defined as four to nine menstrual cycles per year or cycle length greater than 35 days [17] and amenorrhea was defined as at least 3 months without a period [18]. In this study based on Rotterdam criteria (the presence of two or more of the following: Oligo/amenorrhea, clinical and/or biochemical hyperandrogenism, and polycystic ovaries), the PCOS women were selected and those with the menstrual interval being at least 60 days during the last year were included in the study.

Exclusion criteria, on the other hand, were thyroid dysfunction, diabetes mellitus, elevated FSH, hyperprolactinemia, anemia, and coagulation disorders.

Also, the metformin and emmenagogue users (chemical or herbal), pregnant, and lactating women were excluded from the study.

Then, we excluded those who got pregnant, took hormonal medication, or metformin during follow-up (n=2). After explaining the conditions of the study for qualified participants, written consent was obtained from the participants.

Study design

The study protocol was approved by the Ethics Committee of Shahid Beheshti University of Medical Sciences (license No. IR.SBMU.REC. 1395.34) and was registered in the Iranian Registry of Clinical Trials (No. IRCT2016062728664N1). The study was conducted following the guidelines of The Declaration of Helsinki. At initiation of the study pregnancy test, complete blood count (CBC), follicular stimulating hormone (FSH), luteinizing hormone (LH), thyroid-stimulating hormone (TSH), Prolactin, and fasting blood sugar (FBS) were performed to identify those that met the inclusion criteria. In this study, the effect of wet-cupping on menstruation (volume, accuracy, and duration) was a primary outcome. Meanwhile, the effect of excessive body hair, insulin resistance, and Free Androgen Index was a secondary outcome.

Wet-cupping was done on day 26 of the menstruation cycle. The intensity of pain was asked based on the Visual Assessment Scale (VAS) [19] and rated from 1 to 10, with the lowest score indicating the least pain intensity.

Participants were followed up through weekly phone calls and were examined two times after 4 and 12 weeks. Wet-cupping sites were examined in each visit. A comprehensive questionnaire including demographic, reproductive, familial, and past medical history (especially the use of metformin) was filled up for all women after obtaining written consent. Menstrual interval and blood loss were recorded based on Pictorial Blood Loss Assessment [20].

The Ferriman-Gallwey score (FGS) is a known method of evaluating and quantifying hirsutism in women. FGS was assessed using the Ferriman-Gallwey [21] score at baselines and follow-ups. Hormonal and biochemical assessments including FBS, Fasting Insulin, Total Testosterone, and SHBG were conducted before wet-cupping as baseline laboratory data and the end of the study (12 weeks after intervention).

Wet cupping procedure

Participants were invited to do cupping on day 26 of the menstruation cycle. Menstruation was induced by hormonal agents in women who had no menstruation cycle. At first, the antecubital blood sample was obtained in the fasting state and separated serum was frozen and kept at −80 °C until the end of the study for further hormonal and biochemical assessments. Then, patients were prepared for cupping after breakfast. They were asked to sit down on a chair near the bed.

Afterward, the calf muscle area was disinfected using Povidone-iodine and suctioned by a vacuumed cup adjoining to an electrical suction. Negative pressure was created for about a few minutes. Next, multiple superficial oblique scratches with a length of 2–3 mm were made on the hyperemic skin once.

Blood was drawn through the vacuum of the cup, which was placed on the skin. In the next step, the cup was removed and the extracted blood was gently cleaned. This process (only suction and extracting capillary blood) was done in triplicate in the same site. Finally, the location of cupping was covered by a layer of honey and was dressed in gauze. Then, the same process was repeated for the other leg. During the process, whenever the participant felt discomfort, she could lie down on the bed.

Measurements

FBS was measured using the glucose oxidase method (Glucose kit; Pars Azmun, Tehran, Iran), in which both inter-assay and intra-assay coefficients of variation (COV) were 2.2%.

Serum insulin concentration was measured using the ultrasensitive enzyme-linked radioimmunoassay method (Mercodia, Uppsala, Sweden) with a covariance of less than 4%. Circulating SHBG was measured by the Immunoenzymometric Assay method (IEMA kit, Diagnostic Biochem Canada Inc., Ontario, Canada). The assay sensitivity was 10, 0.1 mIU/L, and 0.1 nmol/L, respectively. Also, the intra-assay COVs were 7.3, 6.5, and 7.9%, respectively. Circulating levels of total testosterone were determined by the Enzyme Immuno Assay method (EIA kit, Diagnostic Biochem Canada Inc.). The assay sensitivity and the intra-assay COV were 0.076 nmol/L and 7.6%, respectively. Insulin resistance (IR) was estimated by the homeostasis model assessment (HOMA) through the following formula: HOMA-IR = (Fasting insulin level (mU/L) × FBS (mmol/L)/22.5.

Also, FAI (Free Androgen Index) was estimated as FAI = Total testosterone (nmol/L)/SHBG (nmol/L) × 100

Data analysis

The sample size was estimated via the results of a pilot study. By p=75%, which shows the prevalence of menstrual interval improvement as the primary outcome, power 90% type-I error 5 and 10% dropout, the sample size was estimated 66 participants through the following formula:

$$N = \frac{\left\{ z_{1-\frac{\alpha}{2}} \sqrt{[\pi_{\text{Known}}(1-\pi_{\text{Known}})]} + z_{1-\beta}\sqrt{[\pi_2(1-\pi_2)]} \right\}^2}{\delta^2}$$

Data were described as frequency (percentage) for categorical variables and median (interquartile range) for continuous variables. The normality assumption was checked via the Kolmogorov-Smirnov and Shapiro-Wilk test. To compare values before and after the intervention, paired t-test and Wilcoxon non-parametric statistical tests were applied. Also, for categorical data analysis, the Chi-squared test was considered. p-values<0.05 was considered significant. Statistical analysis was performed using the SPSS software version 16.0.

Results

Figure 1 shows the flow chart of the present study. Of 268 women who were referred to the gynecology clinic, most of them had shorter cycle lengths or were not satisfied with cupping, and 202 were excluded from the study. Eventually, 66 who met the inclusion criteria were invited for cupping. Among these women, 65 women completed the follow-up first visit and 62 completed the follow-up last visit.

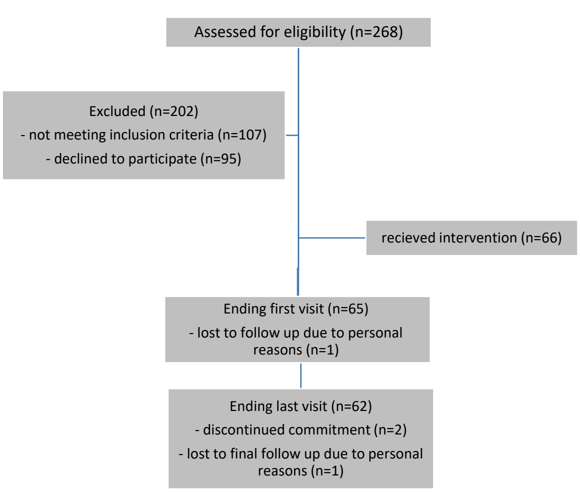

Figure 1: Study flowchart.

Table 1 shows the baseline characteristics of the study population. The mean (±SD) age and interval of menstrual cycles of participants were 29.5 (±4.1) years and 110.69 (±49) days, respectively.

Eighteen patients (27%) suffered from amenorrhea and 48 women (73%) suffered from oligomenorrhea during the last year.

Forty-seven (59.5%) participants were hirsute and 40 (63.5%) participants were identified as IR according to the cutoff value of 2.63 for IR. Also, 30 (47.6%) participants were as hyperandrogenism according to the cutoff value of 3.94 for hyperandrogenemia. The mean excreted blood through wet-cupping was 54(±40) mL.

Table 2 compares the baseline and follow-ups conditions of study participants. After wet-cupping, FGS declined significantly and remained stable until the end of the study (p≤0.001). The frequency of menstruation through three months increased from 0.92 (±0.51) at the start of the study to 1.29 (±1.09) at end of the follow-up

Table 1: Basic characteristic of participants.

Continues variable	**Mean (±SD)**
Age, years	**29.5 (±4.1)**
Onset of disease, years	**14.6 (±7)**
Menstrual cycle, days	**110.69 (±49)**
Mean blood cupping, mL	**54.1 (±40.2)**
BMI, kg/m^{2} [a]	**27.87 (±5)**
Categorized variable	**n (%)**
BMI, kg/m^2	
<25	**20 (30.3%)**
25–29.9	**28 (42.4%)**
≥30	**18 (27.3%)**
Pattern of menstruation	
Amenorrhea	
≥6 month	**18 (27%)**
Oligomenorrhea according to the menstrual interval	
3–6 month	**13 (20%)**
2–3 month	**16 (24%)**
2 month	**19 (29%)**
Efficacy of metformin	
Not effective	**24 (36.3%)**
Effective	**28 (42.4 %)**
Never used	**14 (21.5 %)**
HOMA-IR [b]	
<2.63	**23(36.5%)**
≥2.63	**40(63.5%)**
FAI [c]	
<3.94	**33(52.5%)**
≥3.94	**30(47.6%)**

[a]Body Mass Index, [b]HOMA-IR index = FBS (mmol/L) × Fasting Insulin (μu/mL)/22.5, [c]Free Androgen Index = Total testosterone (nmol/L)/SHBG (nmol/L) × 100.

(p=0.001) with no significant changes in vaginal blood volume (p=0.154). Also, subgroup analysis demonstrated that increasing the frequency of menstrual cycles was only statistically significant for those in the oligomenorrhea group (from (p≤0.01). Excluding those who never used metformin revealed that the chance of improvement of the menstrual cycle by wet cupping was increased by 3.3 times (95.5 CI:2.1–4.5, p=0.036) in those having a previous positive response to metformin compared to those with no response (Table 3). There was no statistically significant change in metabolic, biochemical, and hormonal profiles of participants after wet cupping (Table 2).

Fifty-nine patients (89.3%) were satisfied with wet cupping. Based on the Visual Assessment Scale (VAS), the mean pain score was 5.1 (±2.4) and the intensity of pain was reported as moderate by 42 participants (63.6%).

About 33% of participants (n=22) suffered from mild temporary discomforts at the time of wet-cupping including nausea [(n=15 (22.7%)], vertigo [n=9 (13.6%)], calves weakness [n=5 (7.5%)], headache [n=2 (3%)], vomiting [n=1 (1.5%)], and syncope [n=1 (1.5%)]. These complaints were resolved within a few minutes of rest. Six participants (9%) reported long-term side effects, such as leg weakness in cold weather, which continued for several weeks.

Discussion

To the best of our knowledge, this clinical is the first trial study on only wet cupping on calf muscles in PCOS women. The results show that this process leads to favorable changes in clinical manifestations including improvement of menstruation interval and FGS. The menstruation interval improvement was more prominent in the oligomenorrhea group so no disorder was developed in them.

Wet cupping in Persian Medicine consists of the sucking step (negative pressure is applied to the skin surface using a cup for the creation of hyperemic skin), the scarification step (multiple superficial and subtle incisions on hyperemic skin causing capillary bleeding and peripheral nervous terminal stimulating), and again skin suction (negative pressure cup is applied to allow a longer duration of pressure-dependent filtration to provide better filtration, excessive substance removal, and blood excretion [8, 22].

Table 2: Comparison clinical and biochemical characteristics before and after the intervention.

Continues variable		Baseline data	After intervention			
			4 weeks	Mean diff. (95% CI), p-Value[a]	12 weeks	Mean diff. (95% CI), p-Value[a]
Frequency of menstruation through 3 months	Both groups	0.92 (±0.51)	–	–	1.29 (±1.09)	0.37 (0.13, 0.51), 0.001*
	Amenorrhea	0.23 (±0.19)			0.47 (±0.87)	0.24 (−0.10, 0.31), 0.159
	Oligomenorrhea	1.17 (±0.34)			1.58 (±1.02)	0.41 (0.25, 0.64), 0.002*
Menstrual length		5.96 (±0.25)	–	–	5.69 (±0.26)	−0.27 (−0.32, 0.11), 0.326
FGS		8.6 (±6)	6.1 (±5)	−1.9 (−2.5, −0.5), <0.001*	7.2 (±5)	−1.4 (−2.1, −0.8), <0.001*
Hormonal & biochemical assays (n=48)						
FBS, mg/dL		95.0 (±15)	–	–	92.5 (±12)	−1.5 (−2.9, 1.7), 0.134
Fasting insulin (μu/mL)		16.0 (±9)	–	–	15.9 (±9)	−0.1 (−0.7, 0.7), 0.878
HOMA-IR[c]		3.8(±2.4)	–	–	3.6 (±2)	−0.2 (−0.5, 0.4), 0.984
Testosterone, ng/mL		0.40 (±0.2)	–	–	0.41 (±0.22)	0.1 (−0.5, 0.3), 0.797
SHBG, nmol/L		46.4 (±35)	–	–	44.8 (±42)	−1.6 (−2.5, 1.9),0.862
FAI[d]		4.9 (±3.8)	–	–	4.8 (±4.1)	−0.1 (−0.4, 0.2), 0.595
Categorical variables						
Uterine bleeding volume (n=52)						
Mild		@14 (27%)	–	–	15 (29%)	0.154[b]
Moderate		30 (58%)	–		23 (44%)	
Sever		8 (15%)	–		14 (27%)	

*p-value is significant at the <0.01 level, [a]All values compared to baseline values using Wilcoxon non-parametric and paired t-test in case of normally distributed variable, @n (percentage), [b]obtained from chi-square test, [c]HOMA-IR index = FBS (mmol/L) × Fasting Insulin (μu/mL)/22.5, [d]FAI is calculated using this equation: (FAI = Total testosterone (nmol/L)/SHBG (nmol/L) × 100.

Table 3: Improvement of the menstrual cycle in a group of Metformin users.

		Efficacy of metformin		Odds ratio(95% CI), p-Value[a]
		No effect	Effective	
Were menstrual cycles improved?	No	13 (54.2%)	7 (25%)	3.3 (95% CI:1.2, 4.1), 0.036*
	Yes	11 (45.8%)	21 (75%)	

[a]p-value obtained from fisher exact test; *Significance level <0.05.

A limited number of studies have assessed the effect of wet cupping as a treatment/supplementary modality in PCOS women. The study of Parveen et al. supports our findings of regulating menstrual cycles. In this randomized open-label study, 40 PCOS women with the age range of 18–40 years were assigned into two groups. The first group received herbal medication along with two times wet-cupping on calf muscles before the expected date of menstruation for three consecutive cycles and the second group only received herbal drugs. They observed improvement in menstrual cycles and hormonal profiles including LH & FSH, despite no significant changes in ovarian size [23]. Also, Begum reported regulation of menstruation and normalization of ovaries sizes after 8 times of wet-cupping on calf muscles in a 28 years old PCOS woman with cycle interval >4 months and bilateral polycystic ovaries [24].

Despite the long use of wet-cupping in many countries around the world, the mechanisms through which wet-cupping might be proven as a treatment are unknown. Various studies [25, 26] suggest that the mechanism of wet-cupping is dominated by neural and hematological factors. The first group includes i) regulation of neurotransmitters and hormones and ii) the negative charge of neuronal cells. The latter includes i) regulating the coagulation and anticoagulation systems, ii) decreasing the Hematocrit, ferritin, and causative pathological substances (CPS) [27], and iii) immune system functioning, which is the immune system by making an artificial local inflammation and then activating the complementary system [28].

According to Avicenna's view, the uterus is aligned parallel with calf muscles [8], which means they have a common source in bleeding, nervous system, fascia, etc. Nowadays, it is known that the skin of calf muscles (dermatome) is supplied by a sensory nerve root in the spinal cord, which is associated with uterine afferent fibers [29, 30].

In this study, we supposed that wet-cupping may improve menstruation in three ways. The first way is through stimulating nervous terminals that have a common root with uterine nerves. It is noteworthy that the gastrocnemius area is one of the acupuncture points that is induced in PCOS women [31]. Second, wet-cupping may improve insulin resistance because of binary clinical effects on menstruation and excessive body hair. Studies reported an increase in serum ferritin levels in women with PCOS, especially when glucose tolerance is abnormal [32, 33]. Daniali had already explained that the amount of ferrous in a blood sample obtained from cupping is significantly more than that of a venous blood sample (p=0.022) [34]. On the other hand relationship between acupuncture and insulin sensitivity had been discussed previously [35, 36]. These points may confirm our hypothesis. Also, metformin as an insulin-sensitizing medication has been suggested for the improvement of clinical features and endocrine profiles in women with PCOS. In our study, improvement of menstruation cycles was better in those who had a previous positive response to metformin than those with no response. The group which had already been treated with metformin was cured better in this study and this matter reinforces our hypothesis. Although we have not observed any improvement in hormonal profiles after 12 changes in clinical manifestation reduce after this time.

Third, it has been reported that oxidative stress increases PCOS and insulin resistance [37–39]. Moreover, some studies have shown that wet-cupping removes oxidative stress and lowers oxidative stress [9, 40, 41]. Therefore, it is inferred that wet-cupping can improve the health condition of these patients by dealing with these factors [42, 43].

Our study has several strong points. Chiefly, it is the first quasi-experimental study conducted on PCOS women to assess both the clinical and biochemical effects of wet cupping. Also, in our study, unlike other studies, the intervention was only wet-cupping on the calf muscle without any herbal drug usage.

However, our study has some limitations as well. Since we could not design a placebo procedure like wet cupping, this study was conducted in a quasi-clinical non-randomized mode before and after the trial. Also, PCOM morphology in sonography was not assessed after the intervention. Wet cupping has not been repeated in a specific interval. As a result, some of its temporary effects have not been precisely assessed after three months. We used HOMA-IR as a surrogate marker for assessing the IR. Despite a good correlation between HOMA-IR and gold standard clamp methods [44], it might be inaccurate in PCOS subjects [45].

We concluded that in addition to the necessity of conducting more studies on this procedure and mechanism from different aspects, wet-cupping on calf muscles can be propounded as an optional treatment of PCOS for those reluctant to use chemical medication.

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Ethical approval: The Ethics Committee of Shahid Beheshti University of Medical Sciences deemed and approved the this study (license No. IR.SBMU.REC.1395.34).

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